

BISWA BHUSANA MAHAPATRA

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Bengaluru, India

EDUCATION

Doctor of Philosophy | 2018 - 2024

Ashoka Trust for Research in Ecology and the Environment (ATREE), Manipal Academy of Higher Education (MAHE)

Thesis title: Identifying potential drivers of invasion and assessing the extent of niche shift in globally invasive vertebrates

Integrated M.Sc. (Life Science) | 2013 - 2018

National Institute of Technology Rourkela

Thesis title: Study of *Salmonella* mediated dauer formation in *Caenorhabditis elegans*

EXPERIENCE

Research Integrity Officer (Scientific Officer C) | 2022 - 2025

National Centre for Biological Sciences (NCBS), Tata Institute of Fundamental Research (TIFR)

- Consulting the faculty members, researchers and students on the institutional data integrity policies
- Contributed to developing Standard Operating Procedures (SOPs) and policies
- Documenting all the publications and relevant data
- Maintenance of logistics and regulations of the office
- Procurement and renewal of text integrity check tools
- Assisting in research misconduct investigation
- Teaching research data management to the graduate students of NCBS, inStem & TIGS
- Assisting the Dean's office in preparing institute publication related analyses

PhD Candidate | 2018 - 2024

Ashoka Trust for Research in Ecology and the Environment (ATREE)

- Conducted experiments using GIS, species distribution modelling, and other analytical techniques
- Analysed and interpreted data
- Mentored under graduate student interns
- Drafting reports, research articles
- Presenting research papers in conferences

Master's Student | 2017 - 2018

National Institute of Technology Rourkela

- Prepared mutant strains of *Salmonella* by one step gene inactivation using λ -red recombinase system
- Techniques: DNA isolation, PCR, PCR purification, Gel doc, pharyngeal pumping, dauer assay
- Tools used: ImageJ, R Studio
- Student Mentor for the first year batch (Life Science) through Student Counselling Program

CERTIFICATES

- NRDMS - DST Summer School on Geospatial Technologies (L1) at the National Institute of Technology Karnataka, Surathkal, 2019, July 11-31
- Foundations: Data, Data, Everywhere by Google on Coursera
- Ask Questions to Make Data-Driven Decisions by Google on Coursera
- National Creativity Aptitude Test 2015: Secured All India 89.8 percentile in category 2 (round 1)
- National Creativity Aptitude Test 2017: Secured All India 98 percentile in category 4 (round 1)
- Reviewer certificate for reviewing an article by Global Change Biology

PROJECTS

Identifying medicinal plants for restoring degraded agricultural lands in India

2021

- Estimated the suitable medicinal plants for the degraded lands considering both climate change and socio-economical challenges
- Present and discuss the outputs with different stakeholders of the project
- The work was presented at COP26, Glasgow, England

Collating mammal distribution data for India

2020 - 2021

- Compiling mammal distribution data from literature, ZSI fauna series, and other secondary database
- Geocoding of location data using Python and QGIS
- Compare the mammal distribution map against the IUCN range map

Developing Human footprint Index for India

Summer 2017

- The HFI was estimated using grid based analysis and
- A map was prepared to show gradient of lower HFI to higher HFI across India

Developing Conservation values for amphibians of the Western Ghats of India

Summer 2016

- Preparing distribution map of amphibians in the Western Ghats

PUBLICATIONS

- Bhattacharyya S, **Mahapatra BB**, Palakodeti D, Ladher R, Sane SP & Ramanathan A, Developing institutional publication data archive to maintain reproducibility, transparency and accountability in scientific research: A case study from India. *Insights: the UKSG journal* (Under review)
 - Liriina K, **Mahapatra BB**, Gopal A, Aravind NA, Ratnam J, Sankaran M, Distribution patterns of co-dominant savanna tree species under current and future climates. *PLOS One* (Under review)
 - Aravind NA, Bhat P & **Mahapatra BB**, Niche shifts, low haplotype diversity and invasion potentials of invasive snail *Lissachitina fulica* (Gastropoda: Achatinidae). *Biodiversity & Conservation* (Under review)
 - **Mahapatra BB**, Ravikanth G & Aravind NA, Potential distribution and niche dynamics of invasive mammals. *Ecological Complexity* (Under review)
 - Bhattacharyya S, **Mahapatra BB**, Palakodeti D, Ladher R, Sane SP & Ramanathan A, (2025) General process and specific challenges to build a multi-institutional research integrity office in India. *Ethics and Integrity in Educational Context*, Davis M, Baptista C (Editors). *Springer Nature* (Accepted)
 - **Mahapatra BB**, Das NK, Jadhav A, Roy A & Aravind NA, (2025) Global freshwater mollusc invasion: pathways, potential distribution, and niche shift. *Hydrobiologia*, 852, 1431-1450
 - **Mahapatra BB**, Ravikanth G & Aravind NA, (2024) Do invasive tetrapods conserve their climatic niches? preprint: <https://doi.org/10.1101/2024.03.28.587128> *bioRxiv*
 - **Mahapatra BB** & Aravind NA, (2021) *Laevicaulis haroldi* (Veronicellidae: Gastropoda), a potential future invader to India. *Current Science*, 120, 1781-1785
 - Seetha S, Sahu S, **Mahapatra BB** & Mishra M, (2017) Post embryonic changes in the eye of an economic mango plant pest *Amritodus atkinsoni* Leth. (Hemiptera: Membracoidea: Cicadellidae). In: *Oral A, Bahsi Oral Z (eds) 3rd International Multidisciplinary Microscopy and Microanalysis Congress (InterM), Proceedings, Springer International Publishing*.
 - **Mahapatra BB**, Marathe SA, Meyer-Rochow VB & Mishra M, (2016) A closer look at the feather coloration in the male purple sunbird, *Nectarinia asiatica*. *Micron*, 85, 44-50
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CONFERENCES & WORKSHOPS

- Organised a workshop “*Know your field through pixels*” at SCCS - Bengaluru 2024, Oct 22
- Short talk on “*Freshwater molluscs invasion*” in Molluscs of SouthAsia 2023, ATREE-IISc, Bengaluru, India, Dec 12-13
- Student plenary talk “*Do invasive vertebrate conserve their climatic niches?*” in BES Macro 2023, University of Birmingham, UK, July 12-13 (virtual)
- Speed talk on “*Does invasive mammals follow niche conservatism?*” in SCCS - NY 2022, Oct 3-7 (virtual)
- ePoster on “*Potential distribution and extent of niche shift in invasive mammals across the globe*”, BES and SFEE Annual Meeting: Ecology Across Borders, Dec 12-15 (virtual)
- Talk on “*Distribution and determinants of animal invasive species*” in ATBC 2021, July 21-23 (virtual)
- Participated in “*Workshop on Open Data Kit (ODK) for Field Researchers*” organised by the Academy for Conservation Science and Sustainable Studies, ATREE, 2018, Dec 12-14
- Participated in “*Workshop on Statistical Methods and R Programming for Biologists*” at Indian Statistical Institute Kolkata, 2018, March 7-13
- Participated in “*Meeting on Cell Biology of Infection*” at NCBS - TIFR, 2016, Oct 13-14

INVITED TALKS

- Talk on “*Publication ethics, authorship & open access publishing*” at Karnataka Science and Technology Academy, Bengaluru 2025, Jun 21
- A session on “*Research Integrity and Ethical Leadership*” for PRISM fellowship students and DST sponsored interns at NCBS (invited by the Science Education and Outreach Office) 2025, May 31
- Talk on “*Publication ethics, research misconducts, authorship & open access publishing*” at Karnataka Science and Technology Academy, Bengaluru 2024, Nov 28
- Talk on “*Using text integrity software effectively*” at Academic Integrity & Regulatory Compliance in Life Sciences, iBRIC - inStem 2024, Feb 12-13
- Panelist in Generative AI and its role in contract cheating, Turnitin 2024, Dec 14 (virtual)
- Talk on “*Publication ethics, violation of publication ethics & authorship*” at Karnataka Science and Technology Academy, Bengaluru 2023, Oct 6
- “*Training on Ecological Niche Modelling for scientists of ICFRE, Dehradun*” organised by CES, IISc, 2021, Dec 6-10 (Lecture + Hands on, GIS and MaxEnt SDM)
- “*Workshop on Basics of GIS*” at IIBM Guwahati, organised by ATREE 2020, Feb 12-15 (Lecture on GIS + Hands on training of QGIS)

REFERENCES

Dr. Aravind N.A.

Senior Fellow (Professor)

Ashoka Trust for Research in Ecology and the Environment (ATREE),
Bengaluru 560064, India

Dr. G. Ravikanth

Senior Fellow (Professor)

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Bengaluru 560064, India

Dr. Sabuj Bhattacharyya

Scientist

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Bengaluru 560065, India

Prof. Uma Ramakrishnan

Professor & Dean of Faculty

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Bengaluru 560065, India

Dr. Vidya Devi Negi

Assistant Professor

Indian Institute of Science Education and Research Mohali (IISERM)
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